Sharp Total Variation Delocalization of Bias for the Unadjusted Langevin Algorithm on Gaussian Targets

Simon Lee
University of California, Los Angeles

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Abstract

The unadjusted Langevin algorithm (ULA) with step size h targets $\pi \propto e^{-V}$ but converges to a biased stationary law $\pi_h \neq \pi$. Recent work has shown that this bias *delocalizes*: the error in any k-coordinate marginal grows with k rather than with the full dimension n. This was proven in Wasserstein distance (Chen, Cheng, Niles-Weed & Weare [1]) and in relative entropy at rate $O(hk)$ (Lacker & Zhou [4]), in both cases as upper bounds only. We give the first total variation delocalization results, with matching lower bounds, for Gaussian targets $\pi = \mathcal{N}(0, A^{-1})$ with $mI \preceq A \preceq MI$. We (i) compute the exact stationary covariance Σ_h; (ii) prove the marginal KL is two-sided, $\Theta(h^2 k)$, strictly sharper than the general $O(hk)$; (iii) prove the marginal TV is two-sided, $\Theta(\min(h\sqrt{k}, 1))$; and (iv) identify the exact limiting TV profile $2\Phi(a\tau/(4\sqrt{2})) - 1$ as $h \to 0$, $k \to \infty$ with $h\sqrt{k} \to \tau$, in the isotropic case. All results hold for the full well-conditioned class, with no sparsity or banding assumption.

1 Introduction

For a target $\pi(x) \propto e^{-V(x)}$ on $\mathbb{R}^n$, ULA [7] is the Euler–Maruyama discretization of the overdamped Langevin diffusion $dX_t = -\nabla V(X_t)\,dt + \sqrt{2}\,dB_t$:

$$X_{t+1} = X_t - h\nabla V(X_t) + \sqrt{2h} Z_t, \qquad Z_t \overset{\text{iid}}{\sim} \mathcal{N}(0, I). \tag{1}$$

For fixed $h > 0$ the chain is ergodic with a stationary law $\pi_h \neq \pi$; the discrepancy is the discretization bias. Writing π^u, π_h^u for the marginals on a coordinate set $u \subseteq [n]$, $|u| = k$, the question is how the marginal bias scales in h and k.

Although the full-dimensional bias grows with n, the marginal bias grows only with k; this phenomenon is known as *delocalization*. Chen et al. [1] proved it in a bespoke W_{2,ℓ^∞} metric, $\max_{|u|=k} W_2^2(\pi_h^u, \pi^u) = O(hk \log n)$, and Lacker & Zhou [4] removed the log and upgraded to relative entropy, $\mathrm{KL}(\pi_h^u \| \pi^u) \leq C\, hk$, via a hierarchical entropy recursion. Both are upper bounds only.

Total variation (TV) is the natural metric for hypothesis testing: it quantifies the best achievable test error, is geometry-free, and survives heavy tails where W_2 degenerates. Full-dimensional TV bounds for ULA are well developed [2, 3] (see also [5] for Euler schemes with multiplicative noise), but none of these contains a marginal, delocalized statement. The trivial route is Pinsker's inequality: $\mathrm{TV} \leq \sqrt{\frac{1}{2}\mathrm{KL}} \leq \sqrt{\frac{C}{2}}\sqrt{hk}$, but this is $\sqrt{h}$-loose: already in the one-dimensional Ornstein–Uhlenbeck case the true TV bias is $\Theta(h)$, not $\Theta(\sqrt{h})$. The loss has two potential sources: the structural TV $\leq \sqrt{\mathrm{KL}/2}$ conversion itself, and slack in the input marginal KL bound $O(hk)$ relative to the true stationary KL. This paper resolves both in closed form for Gaussian targets.

Contributions. For $\pi = \mathcal{N}(0, A^{-1})$, $mI \preceq A \preceq MI$:

- **Theorem 1:** the exact stationary law $\pi_h = \mathcal{N}(0, \Sigma_h)$ with $\Sigma_h = [A(I - \frac{h}{2}A)]^{-1}$.
- **Theorem 2:** the marginal KL in closed form, and two-sided: $\Theta(h^2 k)$.
- **Theorem 3:** the marginal TV two-sided: $\Theta(\min(h\sqrt{k}, 1))$.
- **Theorem 4:** the exact limiting TV profile in the isotropic case.

All four theorems require only $mI \preceq A \preceq MI$: no sparsity and no banding. The improvement of the KL rate from $O(hk)$ to $\Theta(h^2k)$ reflects a Talay–Tubaro-type cancellation of the leading $O(h)$ term [8], available here because $\nabla^3 V \equiv 0$; it is consistent with, not contrary to, the general bound of [4]. To our knowledge these are the first TV delocalization results, the first matching (two-sided) lower bounds, and the first exact TV profile in this line.

2 Setup and main results

Let $\mathcal{G}(m, M)$ denote the class of symmetric $A \in \mathbb{R}^{n\times n}$ with $mI \preceq A \preceq MI$, $0 < m \leq M$, and target $\pi = \mathcal{N}(0, A^{-1})$. Since $V(x) = \frac{1}{2}x^\top Ax$ gives $\nabla V = Ax$, the ULA recursion is linear:

$$X_{t+1} = (I - hA)X_t + \sqrt{2h}Z_t. \tag{2}$$

We work in the step-size regime $0 < h \leq h_* := \frac{1}{2M}$ throughout (Theorem 1 also states the wider existence range $h < 2/M$). Fix $u \subseteq [n]$, $|u| = k$, and write $S := (A^{-1})_{uu}$. All constants depend only on (m, M). Throughout, $g(t) := t - \log(1 + t)$, Φ is the standard normal CDF, f_k is the χ^2_k density, and $E := S^{-1/2}(D_h)_{uu}S^{-1/2}$, with D_h defined in Theorem 1.

Theorem 1 (exact stationary law)**.** *For $0 < h < 2/M$, the chain has a unique stationary distribution $\pi_h = \mathcal{N}(0, \Sigma_h)$,*

$$\Sigma_h = [A(I - \tfrac{h}{2}A)]^{-1} = A^{-1} + D_h, \qquad D_h := \tfrac{h}{2}(I - \tfrac{h}{2}A)^{-1}, \tag{3}$$

and for $h \leq h_*$, $\frac{h}{2}I \preceq D_h \preceq \frac{2h}{3}I$.

Theorem 2 (marginal KL, closed form and two-sided)**.** *For $h \leq h_*$,*

$$\mathrm{KL}(\pi_h^u \| \pi^u) = \frac{1}{2}\sum_{i=1}^{k} g(\lambda_i(E)) \qquad \textit{and} \qquad \frac{m^2}{36}h^2k \leq \mathrm{KL}(\pi_h^u \| \pi^u) \leq \frac{4M^2}{9}h^2k. \tag{4}$$

Theorem 3 (two-sided TV delocalization)**.** *For $h \leq h_*$,*

$$c_0(m)\min(h\sqrt{k}, 1) \leq \mathrm{TV}(\pi_h^u, \pi^u) \leq \frac{M\sqrt{2}}{3}h\sqrt{k}, \qquad c_0(m) = \tfrac{1}{2}\cdot 10^{-5}\min\!\big(\tfrac{m}{2}, 1\big). \tag{5}$$

Theorem 4 (exact TV profile, isotropic case)**.** *Let $A = aI$, $a \in [m, M]$. If $h \to 0$, $k \to \infty$ with $h\sqrt{k} \to \tau \in (0, \infty)$, then*

$$\mathrm{TV}(\pi_h^u, \pi^u) \longrightarrow 2\Phi\Big(\frac{a\tau}{4\sqrt{2}}\Big) - 1. \tag{6}$$

3 Proofs

3.1 Strategy

Both marginals are centered Gaussians (Theorem 1), so the pair (π_h^u, π^u) is determined, up to an invertible linear map that preserves TV and KL, by the whitened perturbation $E = S^{-1/2}(D_h)_{uu}S^{-1/2}$. Whitening and diagonalizing reduce it to a product of scalar pairs $\mathcal{N}(0, 1+\lambda_i)$ vs. $\mathcal{N}(0,1)$, indexed by the eigenvalues $\lambda_1, \ldots, \lambda_k$ of E, each of order h (Lemma 3.1). The KL is then an exact sum over the λ_i, and Pinsker converts it to the sharp TV upper bound. For the lower bound and the full TV profile, the whitened likelihood ratio is monotone in $|z|^2$, so the TV equals χ_k^2 mass on an interval of width $\Theta(hk)$ about the mean k. Anti-concentration (Lemma 3.2) then turns this into the matching lower bound, and a local limit evaluates it exactly when $A = aI$.

3.2 Proof of Theorem 1

Proof. (Recursion.) If $X \sim \mathcal{N}(0,\Sigma)$ and $X' = (I-hA)X + \sqrt{2h}Z$ with $Z \sim \mathcal{N}(0, I)$ independent, then $X' \sim \mathcal{N}(0, (I-hA)\Sigma(I-hA) + 2hI)$. A centered Gaussian is stationary iff Σ is a fixed point of $\mathcal{T}(\Sigma) = (I-hA)\Sigma(I-hA) + 2hI$.

(Solve.) In the eigenbasis of A, seeking Σ a function of A decouples $\mathcal{T}$ coordinatewise:

$$\sigma_a^2 = (1-ha)^2\sigma_a^2 + 2h \iff \sigma_a^2\, ha(2-ha) = 2h \iff \sigma_a^2 = \frac{1}{a(1-\frac{h}{2}a)}, \tag{7}$$

finite and positive for every $a \in [m, M]$ iff $h < 2/M$; hence $\Sigma_h = [A(I - \frac{h}{2}A)]^{-1}$.

(Additive form.) Since all factors commute, with $W = \frac{h}{2}A$:

$$\Sigma_h - A^{-1} = A^{-1}\big[(I-W)^{-1} - I\big] = A^{-1}(I-W)^{-1}W = \tfrac{h}{2}(I - \tfrac{h}{2}A)^{-1} =: D_h. \tag{8}$$

(Sandwich.) For $h \le h_*$, $0 \preceq \frac{h}{2}A \preceq \frac{1}{4}I$, so the eigenvalues $\frac{h}{2}(1-\frac{h}{2}a)^{-1}$ of D_h lie in $[\frac{h}{2}, \frac{2h}{3}]$.

(Uniqueness.) For any stationary μ, run two chains from $X_0 \sim \mu$ and $Y_0 \sim \mathcal{N}(0, \Sigma_h)$ with shared noise: $X_t - Y_t = (I-hA)^t(X_0 - Y_0)$ and $\|I - hA\|_{\mathrm{op}} = \max_a |1-ha| < 1$, so $X_t - Y_t \to 0$ a.s.; testing against bounded Lipschitz functions (which determine a measure) and dominated convergence give $\mu = \mathcal{N}(0, \Sigma_h)$. $\square$

3.3 Two lemmas

Lemma 3.1 (whitening reduction)**.** *Let $h \le h_*$ and let $\lambda_1 \ge \cdots \ge \lambda_k$ be the eigenvalues of E. Then:*

(i) $\pi^u = \mathcal{N}(0, S)$ *and* $\pi_h^u = \mathcal{N}(0, S + (D_h)_{uu})$*;*

(ii) an invertible linear map carries (π_h^u, π^u) *to* $\big(\bigotimes_{i\le k}\mathcal{N}(0, 1+\lambda_i), \mathcal{N}(0, I_k)\big)$*, so TV and KL coincide for the two pairs;*

(iii) $\frac{mh}{2} \le \lambda_i \le \frac{2Mh}{3} \le \frac{1}{3}$ *for every* i*.*

Proof. (i) Theorem 1 and the marginalization rule for Gaussians. (ii) Apply $x_u \mapsto S^{-1/2}x_u$, sending the pair to $(\mathcal{N}(0, I+E), \mathcal{N}(0, I))$; diagonalize $E = U\Lambda U^\top$ and apply $z \mapsto U^\top z$, under which $\mathcal{N}(0,I)$ is invariant and $\mathcal{N}(0, I+E) \mapsto \mathcal{N}(0, I+\Lambda)$. TV and KL are invariant under invertible affine maps (the Jacobian factors cancel). (iii) By Theorem 1, $\frac{h}{2}I \preceq D_h \preceq \frac{2h}{3}I$; principal submatrices inherit two-sided spectral bounds (test on zero-padded vectors), so the same holds for $(D_h)_{uu}$; conjugating by $S^{-1/2}$ gives $\frac{h}{2}S^{-1} \preceq E \preceq \frac{2h}{3}S^{-1}$. Since $mI \preceq A \preceq MI$ implies $mI \preceq S^{-1} \preceq MI$ (invert, restrict, invert), every eigenvalue lies in $[\frac{mh}{2}, \frac{2Mh}{3}]$, and $\frac{2Mh}{3} \le \frac{1}{3}$ for $h \le h_*$. $\square$

Lemma 3.2 (χ^2_k anti-concentration near the mean)**.** *For $k \ge 16$ and all $|x| \le 2\sqrt{k}$: $f_k(k+x) \ge 10^{-5}/\sqrt{k}$.*

Proof. Let $\varphi = \log f_k$ and $y = x/k$, so $|y| \le 2/\sqrt{k} \le \frac{1}{2}$. Then

$$\varphi(k+x) - \varphi(k) = (\tfrac{k}{2} - 1)\log(1+y) - \tfrac{x}{2} = \tfrac{k}{2}\big[\log(1+y) - y\big] - \log(1+y). \tag{9}$$

Since $|\log(1+y) - y| = g(y) \le 2y^2$ for $|y| \le \frac{1}{2}$, the first term is at most $ky^2 = x^2/k \le 4$ in absolute value, and $|\log(1+y)| \le 0.7$; hence

$$|\varphi(k+x) - \varphi(k)| \le 4.7 \qquad \text{for all } |x| \le 2\sqrt{k}. \tag{$\star$}$$

Chebyshev with $\operatorname{Var}\chi^2_k = 2k$ gives $\mathbb{P}(|\chi^2_k - k| > 2\sqrt{k}) \le \frac{1}{2}$, so $[k - 2\sqrt{k}, k + 2\sqrt{k}]$ carries mass $\ge \frac{1}{2}$; by $(\star)$, $\frac{1}{2} \le e^{4.7} f_k(k) \cdot 4\sqrt{k}$, so $f_k(k) \ge (8e^{4.7}\sqrt{k})^{-1}$, and one more application of $(\star)$ gives $f_k(k+x) \ge (8e^{9.4}\sqrt{k})^{-1} \ge 10^{-5}/\sqrt{k}$. $\square$

Other tools used repeatedly: the elementary bounds $\frac{2}{9}t^2 \le g(t) \le 2t^2$ on $|t| \le \frac{1}{2}$ (from $g(t) = t^2\int_0^1 (1-v)(1+vt)^{-2}dv$), the TV identity $\mathrm{TV}(P,Q) = P(A) - Q(A)$ for $A = \{p > q\}$, Pinsker's inequality [6], the Gaussian KL formula $\mathrm{KL}(\mathcal{N}(0, I+E)\|\mathcal{N}(0,I)) = \frac{1}{2}[\operatorname{tr} E - \log\det(I+E)]$, and the fact that marginalization does not increase TV.

3.4 Proof of Theorem 2

Proof. By Lemma 3.1(ii) and the Gaussian KL formula,

$$\mathrm{KL}(\pi^u_h\|\pi^u) = \tfrac{1}{2}\big[\operatorname{tr} E - \log\det(I+E)\big] = \tfrac{1}{2}\sum_{i=1}^{k} g(\lambda_i(E)). \tag{10}$$

Since every $\lambda_i \in [\frac{mh}{2}, \frac{2Mh}{3}] \subset (0, \frac{1}{3}]$ (Lemma 3.1(iii)) and $\frac{2}{9}t^2 \le g(t) \le 2t^2$ there,

$$\mathrm{KL} \;\ge\; \tfrac{1}{2}k \cdot \tfrac{2}{9}\big(\tfrac{mh}{2}\big)^2 = \frac{m^2h^2k}{36}, \qquad \mathrm{KL} \;\le\; \tfrac{1}{2}k \cdot 2\big(\tfrac{2Mh}{3}\big)^2 = \frac{4M^2h^2k}{9}. \tag{11}$$

$\square$

3.5 Proof of Theorem 3

Proof. Upper bound. Pinsker applied to Theorem 2:

$$\mathrm{TV} \le \sqrt{\tfrac{1}{2}\mathrm{KL}} \le \sqrt{\tfrac{1}{2} \cdot \tfrac{4M^2h^2k}{9}} = \frac{M\sqrt{2}}{3}h\sqrt{k}. \tag{12}$$

Lower bound. By Lemma 3.1(ii) the pair is $P_1 = \bigotimes_i \mathcal{N}(0, 1+\lambda_i)$ vs. $P_0 = \mathcal{N}(0, I_k)$; set $\delta := \lambda_{\min}(E) \in [\frac{mh}{2}, \frac{1}{3}]$.

Case $k \geq 16$. Let $R(z) = |z|^2$. Under P_1, $R \overset{d}{=} \sum_i (1+\lambda_i)\xi_i^2 \geq (1+\delta)\sum_i \xi_i^2$ pointwise for a single $\xi \sim \mathcal{N}(0, I_k)$, so $P_1(R > t) \geq \mathbb{P}(\chi_k^2 > t/(1+\delta))$ for every t. Taking $t = k$,

$$\mathrm{TV} \;\geq\; P_1(R > k) - P_0(R > k) \;\geq\; \mathbb{P}\Big(\chi_k^2 \in \big(\tfrac{k}{1+\delta}, k\big]\Big). \tag{13}$$

This interval has length $\frac{k\delta}{1+\delta} \geq \frac{k\delta}{2}$, and its intersection with $[k - 2\sqrt{k}, k]$ has length $\geq \min(\frac{k\delta}{2}, 2\sqrt{k})$ and lies in Lemma 3.2's window; integrating the density bound,

$$\mathrm{TV} \;\geq\; \frac{10^{-5}}{\sqrt{k}} \min\Big(\frac{k\delta}{2}, 2\sqrt{k}\Big) = 10^{-5} \min\Big(\frac{\delta\sqrt{k}}{2}, 2\Big) \;\geq\; \frac{10^{-5}}{2} \min(\delta\sqrt{k}, 1). \tag{14}$$

Finally, $\min(\delta\sqrt{k}, 1) \geq \min(\frac{m}{2} h\sqrt{k}, 1) \geq \min(\frac{m}{2}, 1)\min(h\sqrt{k}, 1)$, using $\min(ax, 1) \geq \min(a, 1)\min(x, 1)$; this is the claim with $c_0(m) = \frac{1}{2} \cdot 10^{-5} \min(\frac{m}{2}, 1)$.

Case $k \leq 15$. Project to one coordinate $a \in u$ (TV does not increase): the pair becomes $(\mathcal{N}(0, s+d), \mathcal{N}(0, s))$ with $s = (A^{-1})_{aa} \leq \frac{1}{m}$, $d = (D_h)_{aa} \geq \frac{h}{2}$, and whitening gives $(\mathcal{N}(0, 1+\delta_1), \mathcal{N}(0, 1))$ with $\delta_1 = d/s \in [\frac{mh}{2}, \frac{1}{3}]$. The same domination step at $t = 1$:

$$\mathrm{TV} \;\geq\; \mathbb{P}\Big(\chi_1^2 \in \big(\tfrac{1}{1+\delta_1}, 1\big]\Big). \tag{15}$$

The interval lies in $[\frac{3}{4}, 1]$, where $f_1(r) = \frac{e^{-r/2}}{\sqrt{2\pi r}} \geq \frac{1}{5}$, and has length $\frac{\delta_1}{1+\delta_1} \geq \frac{3\delta_1}{4}$, so $\mathrm{TV} \geq \frac{3mh}{40}$. Since $k \leq 15$, $h \geq h\sqrt{k}/\sqrt{15}$, whence $\mathrm{TV} \geq \frac{3m}{40\sqrt{15}} h\sqrt{k} \geq \frac{m}{52}\min(h\sqrt{k}, 1) \geq c_0(m)\min(h\sqrt{k}, 1)$. $\square$

Remark 3.1. The two-sided match identifies the rate $\Theta(\min(h\sqrt{k}, 1))$: the $h\sqrt{k}$ law until saturation at $\mathrm{TV} = \Theta(1)$ once $h\sqrt{k} \gtrsim 1$. The constant 10^{-5} from Lemma 3.2 is far from sharp; Theorem 4 gives the sharp isotropic constant.

3.6 Proof of Theorem 4

Proof. Here $A = aI$, so $\Sigma_h = \sigma^2 a^{-1} I$ with $\sigma^2 = (1 - \frac{ha}{2})^{-1} = 1 + \lambda$, $\lambda = \frac{ha/2}{1 - ha/2}$. Whitening reduces the pair to $(\mathcal{N}(0, \sigma^2 I_k), \mathcal{N}(0, I_k))$.

Step 1 (reduction to a χ^2 interval). The density ratio $\frac{p_1}{p_0}(z) = \sigma^{-k}\exp(\frac{1}{2}(1 - \sigma^{-2})|z|^2)$ is strictly increasing in $R = |z|^2$, crossing 1 at $t_* := k\log\sigma^2/(1 - \sigma^{-2})$, so the optimal event is $\{R > t_*\}$ and

$$\mathrm{TV} = \mathbb{P}(\sigma^2\chi_k^2 > t_*) - \mathbb{P}(\chi_k^2 > t_*) = \mathbb{P}\Big(\chi_k^2 \in \big(\tfrac{t_*}{\sigma^2}, t_*\big]\Big). \tag{16}$$

Step 2 (endpoints). As $\lambda \to 0$: $\log\sigma^2 = \lambda - \frac{\lambda^2}{2} + O(\lambda^3)$ and $1 - \sigma^{-2} = \lambda - \lambda^2 + O(\lambda^3)$, so $t_* = k(1 + \frac{\lambda}{2} + O(\lambda^2))$ and $t_*/\sigma^2 = k(1 - \frac{\lambda}{2} + O(\lambda^2))$. In the fluctuation variable $r = k + v\sqrt{k}$ the endpoints are

$$v_\pm(k) = \pm\tfrac{\lambda\sqrt{k}}{2} + O(\lambda^2\sqrt{k}) \;\longrightarrow\; \pm b, \qquad b := \tfrac{a\tau}{4}, \tag{17}$$

since $\lambda\sqrt{k} \to \frac{a\tau}{2}$ and $\lambda^2\sqrt{k} = (\lambda\sqrt{k})\lambda \to 0$.

Step 3 (local limit for χ_k^2). Fix $\Lambda > 0$. For $|v| \leq \Lambda$, expanding $\varphi = \log f_k$ as in Lemma 3.2 but to second order, with $y = v/\sqrt{k}$:

$$\varphi(k + v\sqrt{k}) - \varphi(k) = (\tfrac{k}{2} - 1)\log(1+y) - \tfrac{v\sqrt{k}}{2} = -\frac{v^2}{4} + O\Big(\frac{|v| + |v|^3}{\sqrt{k}}\Big) \quad \text{uniformly,} \tag{18}$$

so $f_k(k+v\sqrt{k}) = f_k(k)e^{-v^2/4}(1+\varepsilon_k(v))$ with $\sup_{|v|\le\Lambda}|\varepsilon_k| \to 0$. Writing $1 = \sqrt{k}\int f_k(k+v\sqrt{k})\,dv$, the region $|v| > \Lambda$ carries mass $\le 2/\Lambda^2$ (Chebyshev), and $|v| \le \Lambda$ contributes $\sqrt{k}f_k(k)(1+o(1))\int_{-\Lambda}^{\Lambda} e^{-v^2/4}dv$; letting $k \to \infty$ then $\Lambda \to \infty$ yields $\sqrt{k}f_k(k) \to (4\pi)^{-1/2}$. Hence, since $v_\pm \to \pm b$ with the integrand uniformly controlled,

$$\mathrm{TV} = \sqrt{k}\int_{v_-}^{v_+} f_k(k+v\sqrt{k})\,dv \;\longrightarrow\; \frac{1}{\sqrt{4\pi}}\int_{-b}^{b} e^{-v^2/4}dv \;\overset{v=w\sqrt{2}}{=}\; 2\Phi\Big(\frac{a\tau}{4\sqrt{2}}\Big) - 1. \tag{19}$$

□

Remark 3.2. For small τ, $2\Phi(\frac{a\tau}{4\sqrt{2}}) - 1 \approx \frac{a\tau}{4\sqrt{\pi}}$, recovering the linear $h\sqrt{k}$ rate with an explicit constant.

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